

* Particle in a 1-D Box

Infinite potential well

* Approach

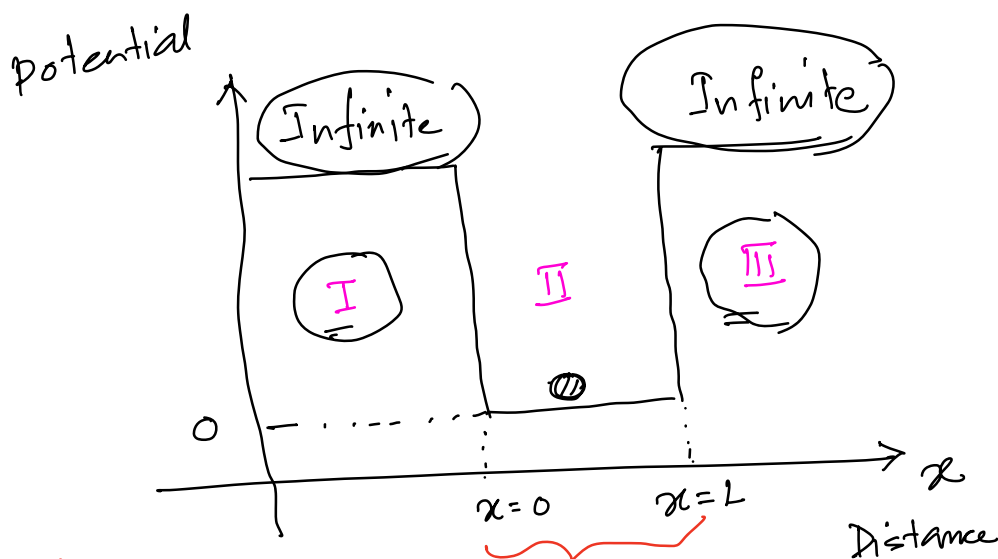
- * Finite potential well
- * tunnelling through finite potential barrier

Region I & II

$$V(x) = \infty$$

Region II

$$V(x) = 0$$



* particle remains confined in region II or in the box

particle is not in R-I

R-I $\boxed{\psi = 0}$

R-II
 $\psi \neq 0$

R-III $\boxed{\psi = 0}$

Schrodinger eqⁿ

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \cancel{V\psi} = E\psi \rightarrow (1)$$

$V(x)=0$

R-II

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + E\psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2mE}{\hbar^2} \psi = 0$$

$$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0 \rightarrow (2)$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

* General solution

$$\psi = A \sin(kx) + B \cos(kx)$$

$$A, B \text{ \& } k = ?$$

* Boundary condition

$\psi \rightarrow$ continuous *

$\frac{\partial \psi}{\partial x} \rightarrow$ continuous *

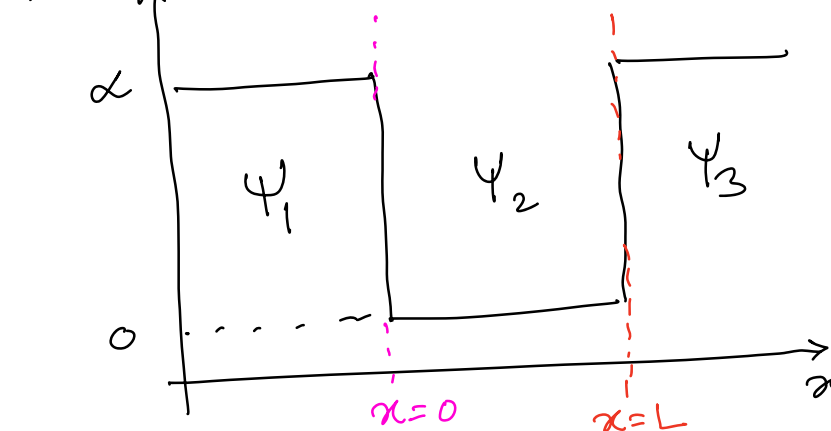
$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1 \quad *$$

Potential

$x=0$ ψ is continuous

$$\psi_1 = \psi_2 \quad | \quad x=0$$

$$\psi_1 = 0$$



$$\psi_2|_{x=0} = \underbrace{A \sin 0} + \underbrace{B \cos 0}_1 = 0 + B$$

$$|\psi_2|_{x=0} = B = \psi_1 = 0$$

$$\underline{B = 0}$$

$$\psi = A \sin kx + \cancel{B \cos kx}^{=0} = A \sin kx$$

$$\underline{\psi = A \sin kx}$$

$$\underline{x=L}$$

$$\psi_2|_{x=L} = \psi_3|_{x=L}$$

$$\Rightarrow A \sin(kL) = 0$$

$$\underline{A = 0}$$

Not possible

$$B=0, A=0$$

$$\underline{\psi = 0}$$

$$\sin kL = 0$$

$$kL = n\pi$$

$$\Rightarrow \boxed{k = \frac{n\pi}{L}}$$

$$\psi = \underline{A} \sin\left(\frac{n\pi}{L} x\right)$$

We know

$$\int_{-\alpha}^{\alpha} \underline{\psi^2} dx = 1$$

$$\int_0^L A^2 \sin^2\left(\frac{n\pi}{L} x\right) dx = 1$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$= A^2 \int_0^L \frac{1 - \cos\left(\frac{2n\pi}{L} x\right)}{2} dx = 1$$

$$= \frac{A^2}{2} \left[\int_0^L dx - \int_0^L \cos\left(\frac{2n\pi x}{L}\right) dx \right] = 1$$

$= 0$

$$\int_0^L \cos\left(\frac{2n\pi x}{L}\right) dx = \left[\frac{\sin \frac{2n\pi x}{L}}{\frac{2n\pi}{L}} \right]_0^L$$

$$= \frac{L}{2n\pi} [\sin 2n\pi - \sin 0]$$

$$= 0$$

$$\Rightarrow \frac{A^2}{2} L = 1$$

$$\Rightarrow A = \sqrt{\frac{2}{L}}$$

$$\Psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L} x\right)$$

→ ③

① Sine wave

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$k = \frac{n\pi}{L}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

$\hbar, \pi, 2, m$
constant

$L \rightarrow$ you know
= $\alpha=0$ to $\alpha=L$

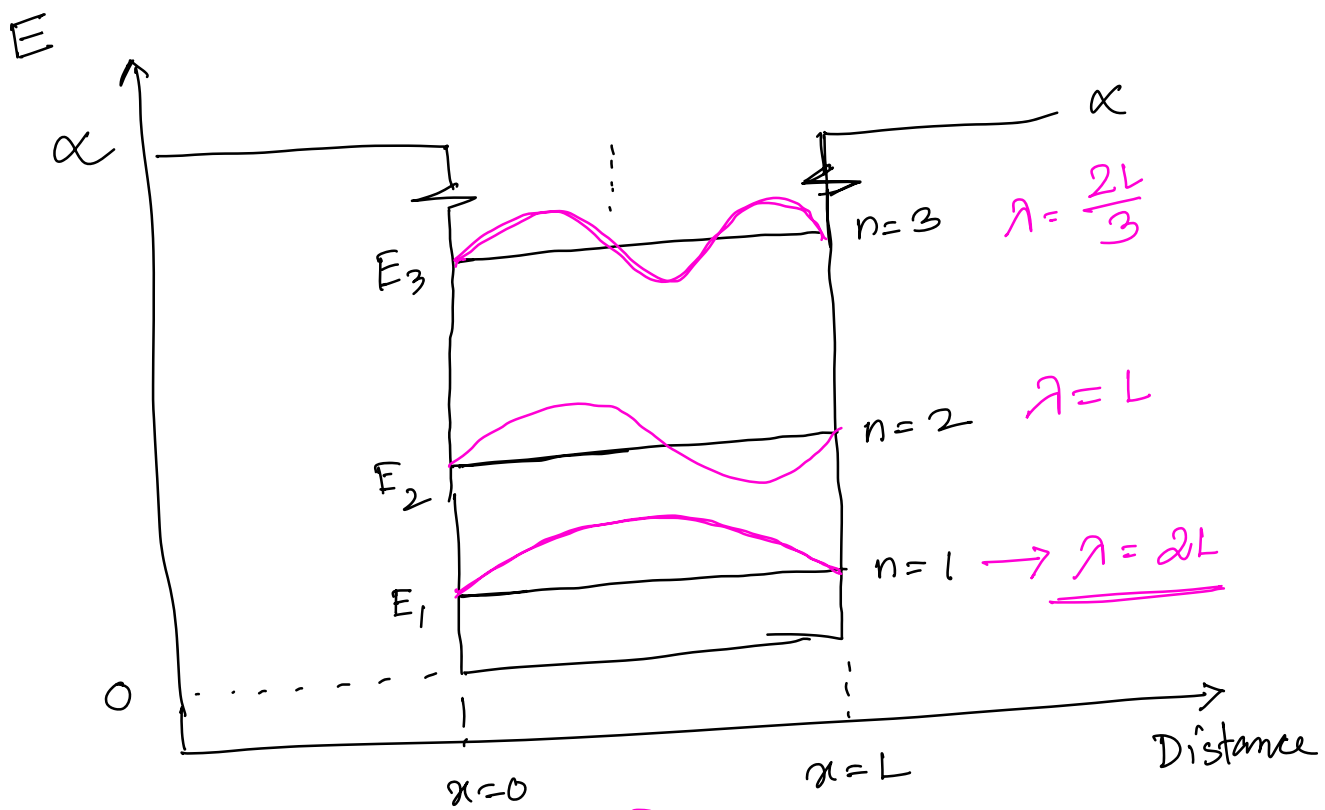
$$E_n \propto n^2$$

1, 2, 3, ...

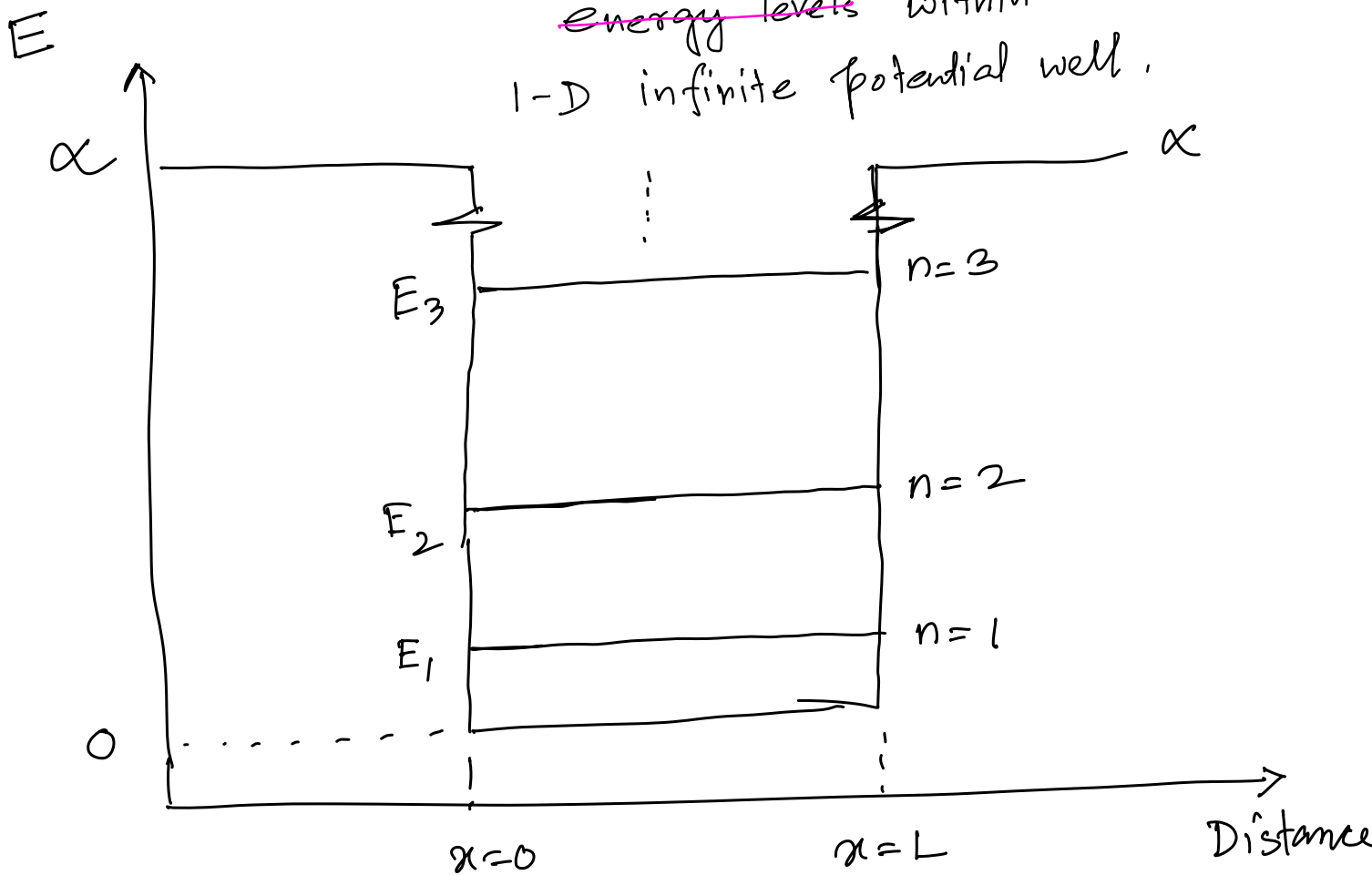
$$k = \frac{n\pi}{L} = \frac{2\pi}{\lambda}$$

$$k = \frac{2\pi}{\lambda}$$

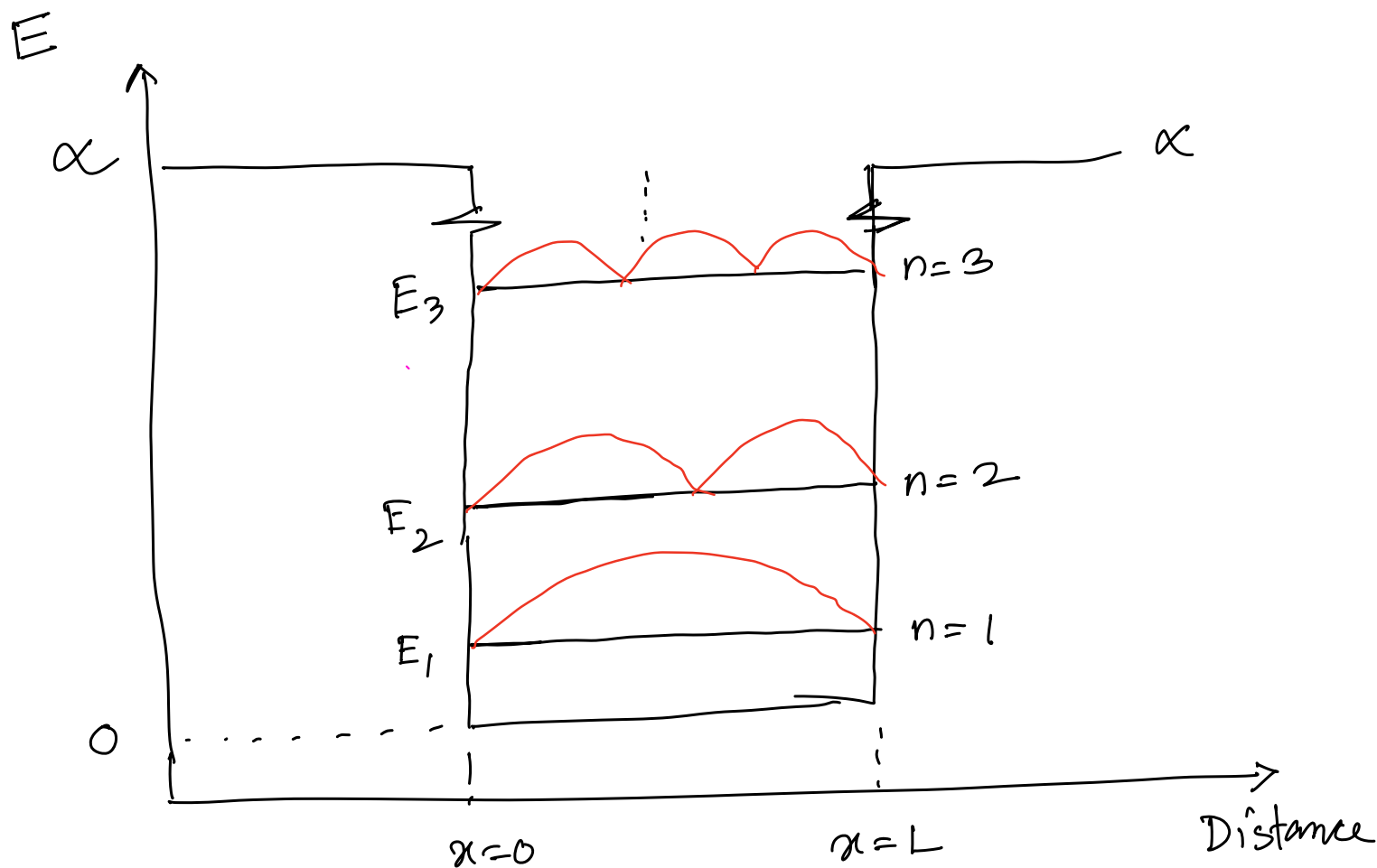
$$\lambda = \frac{2L}{n}$$



wave fn
~~Energy levels~~ within
 1-D infinite potential well.



energy levels
~~wave function~~ within
 1-D infinite potential well.



probability: within
1-D infinite potential well.

Schrodinger general eqⁿ

based
on
R-I

R-II

R-III

→ simplify these eq^{ns} based on scenario.

→ General solⁿ.

→ Using boundary condⁿ $\psi \rightarrow$ contd.,
 $\frac{\partial \psi}{\partial x} \rightarrow$ cond.

$$\int_{-\alpha}^{\alpha} \psi^2(x) dx = 1$$

→ Get simplified solutions.